

MA388 Sabermetrics — Lesson Objectives Mapped to Course Objectives

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AY27-1

Course Objectives

CO1: Use the investigative process in statistics to answer research questions of interest in sabermetrics.

CO2: Use statistical software to obtain, prepare, summarize, visualize, and model complex data.

CO3: Identify and employ advanced statistical methods for estimation and prediction to answer questions in sabermetrics.

CO4: Draw appropriate conclusions from statistical analyses and communicate them to general audiences.

CO5: Appreciate the importance and limitations of statistical inquiry to answer questions of interest.

CO6: Be critical consumers of statistically based results and recognize whether reported results reasonably follow from the study and analysis conducted.

Lesson 1 — Course Introduction

#	Lesson Objective	CO
1.1	Identify major baseball data sources (Lahman, Retrosheet, PITCHf/x, Statcast) and their appropriate use cases.	1, 2
1.2	Navigate course resources including texts, problem sets, and the course calendar to plan learning.	1
1.3	Recognize the three pillars of the course: baseball, statistics, and data science skills.	5
1.4	Explore data using basic tidyverse verbs (select, filter, mutate, group_by, summarize, arrange) to calculate home run rates over time.	2
1.5	Produce a line plot showing how home run rates have changed across decades.	2, 4

Lesson 2 — Introduction to R

#	Lesson Objective	CO
2.1	Compile a Quarto Markdown document to PDF using TinyTeX.	2
2.2	Use <code>left_join()</code> and other join operations to merge information across relational database tables.	2
2.3	Use the native R pipe <code> ></code> and tidyverse verbs to answer a question about career batting average leaders.	1, 2

Lesson 3 — Introduction to R Part II

#	Lesson Objective	CO
3.1	Implement the split-apply-combine strategy using <code>group_by/summarize</code> and <code>split()/map_df()</code> .	2
3.2	Write a custom function and apply it across grouped data to find per-season win leaders.	2, 3
3.3	Distinguish between <code>select()</code> and <code>pull()</code> for extracting data from a data frame.	2

Lesson 4 — R Graphics Part I

#	Lesson Objective	CO
4.1	Use <code>ggplot</code> with aesthetic mappings (color, linetype, shape) to visualize data in long format.	2
4.2	Transform data between wide and long format using <code>pivot_wider()</code> .	2
4.3	Create a scatter plot with color-coded groups and meaningful axis labels.	2, 4
4.4	Map a function over a sequence of years to compute cumulative career home run leaders.	2, 3

Lesson 5 — R Graphics Part II

#	Lesson Objective	CO
5.1	Create factor variables using <code>cut()</code> to categorize numeric data by baseball era.	2
5.2	Construct boxplots and line plots with multiple groups (e.g., salary by decade, runs per game by league).	2
5.3	Use <code>geom_smooth()</code> to add smoothed trends to scatter plots.	2
5.4	Evaluate Hall of Fame voting patterns and compare pitcher statistics by induction status using visualization.	1, 4

Lesson 6 — Runs and Wins Part I

#	Lesson Objective	CO
6.1	Fit a linear regression model to quantify the relationship between run differential and wins.	3
6.2	Interpret regression coefficients and derive the “10 runs = 1 win” rule of thumb.	3, 4
6.3	Assess model fit using residual plots and <code>broom::augment()</code> .	3, 6
6.4	Determine the number of wins typically needed to make the playoffs using historical data.	1, 4

Lesson 8 — Runs and Wins Part II

#	Lesson Objective	CO
8.1	Compare three models (linear, Pythagorean with $k=2$, Pythagorean with estimated k) for predicting win percentage.	3, 6
8.2	Derive and compute incremental runs per win under the Pythagorean model.	3
8.3	Use RMSE to evaluate and compare model performance.	3, 6

#	Lesson Objective	CO
8.4	Estimate the Pythagorean exponent via log-log regression without an intercept.	3

Lesson 9 — Run Expectancy Matrix

#	Lesson Objective	CO
9.1	Construct a run expectancy matrix from Retrosheet play-by-play data.	2, 3
9.2	Interpret run expectancy values for each game state (bases/outs combination).	4
9.3	Calculate the run value of individual plays using the change in run expectancy.	3
9.4	Distinguish between a run expectancy matrix and a run probability matrix.	5

Lesson 10 — Value of Hits and Stealing

#	Lesson Objective	CO
10.1	Compute RE24 to aggregate a player's run value across all plate appearances.	3, 4
10.2	Compare batting average and slugging percentage to run-value-based measures of performance.	6
10.3	Evaluate the expected value of a stolen base attempt using the run expectancy matrix.	3
10.4	Calculate mean run values for singles, doubles, triples, and home runs.	3, 4

Lesson 11 — Swinging Tendencies

#	Lesson Objective	CO
11.1	Use regular expressions and stringr functions to process pitch sequence data.	2
11.2	Create indicator variables for whether a plate appearance passed through each ball-strike count.	2
11.3	Compute and visualize the mean run value for each count.	3, 4
11.4	Explain why ball-strike count affects expected run value.	1, 5

Lesson 12 — Umpire Tendencies

#	Lesson Objective	CO
12.1	Collect Statcast pitch-level data using the <code>baseballr</code> package.	2
12.2	Fit a LOESS model to estimate the probability of a called strike as a function of pitch location.	3
12.3	Visualize umpire strike zones across different counts using contour plots and faceting.	4
12.4	Assess whether umpires adjust their strike zone based on the count.	1, 6

Lesson 13 — Catcher Framing Visualization

#	Lesson Objective	CO
13.1	Import and merge Statcast pitch data with player ID maps to add catcher names.	2
13.2	Create faceted visualizations to compare called-strike proportions across catchers.	2, 4
13.3	Discuss challenges in isolating catcher framing ability from confounding factors.	5, 6
13.4	Calculate called-strike proportions for pitches near the edge of the strike zone.	3

Lesson 14 — Logistic Regression

#	Lesson Objective	CO
14.1	Fit a logistic regression model for a binary response (called strike vs. ball).	3
14.2	Interpret logistic regression coefficients in terms of odds and log-odds.	3, 4
14.3	Explain why logistic regression uses the logit link (sigmoid transformation).	3
14.4	Identify confounders (e.g., count) and adjust for them in a multiple logistic regression model.	3, 5

Lesson 15 — Catcher Framing Modeling

#	Lesson Objective	CO
15.1	Fit a generalized additive model (GAM) to model called-strike probability as a smooth function of pitch location.	3
15.2	Use a GAM to adjust for pitch location when comparing catchers.	3
15.3	Compare a simple linear logistic model with a smooth spatial model for strike-zone decisions.	3, 6
15.4	Extend a two-catcher comparison to consider modeling many catchers simultaneously.	3, 5

Lesson 16 — Catcher Framing Random Effects

#	Lesson Objective	CO
16.1	Fit a fixed-effects model with many catcher indicators and discuss its limitations.	3, 6
16.2	Fit a random-effects (multilevel) model to estimate catcher-to-catcher variability.	3
16.3	Use likelihood ratio tests to compare nested models with and without catcher random effects.	3, 6
16.4	Interpret variance components to assess whether catchers meaningfully affect called-strike rates.	4, 5

Lesson 18 — Peak Trajectories Part I

#	Lesson Objective	CO
18.1	Fit a quadratic model to describe a player's OPS as a function of age.	3
18.2	Derive expressions for peak age and peak OPS from a quadratic model.	3
18.3	Discuss the trade-offs of using a simple parametric curve to model career trajectories.	5, 6
18.4	Explain why centering age at 30 improves interpretability of the model coefficients.	3

Lesson 19 — Peak Trajectories Part II

#	Lesson Objective	CO
19.1	Map a quadratic model function over multiple players to compare career trajectories.	2, 3
19.2	Identify which historical home run hitter had the highest estimated peak HR rate.	1, 4
19.3	Recognize unusual trajectory shapes and hypothesize explanations (e.g., injuries, era effects).	5, 6

Lesson 20 — Multilevel Modeling

#	Lesson Objective	CO
20.1	Fit individual quadratic OBP trajectories and identify their limitations (overfitting for short careers).	3, 6
20.2	Fit a Bayesian multilevel model using brms that pools information across players.	3
20.3	Compare individual-model estimates with multilevel-model estimates (shrinkage effect).	3, 6

#	Lesson Objective	CO
20.4	Explain how Bayesian methods combine prior information with observed data to improve estimates.	5

Lesson 21 — Simulation Part I (Markov Chains)

#	Lesson Objective	CO
21.1	Construct a transition probability matrix from Retrosheet play-by-play data.	2, 3
21.2	Model a baseball half-inning as a Markov chain and simulate 10,000 half-innings.	1, 3
21.3	Compute the distribution of runs scored per half-inning and the average runs per half-inning.	1, 4
21.4	Discuss the memoryless property of Markov chains and whether it allows for momentum.	5

Lesson 22 — Simulation Part II (Season Simulation)

#	Lesson Objective	CO
22.1	Use the Bradley-Terry model to compute the probability one team beats another.	3
22.2	Build a Monte Carlo simulation of an entire baseball season, including schedule generation.	1, 2
22.3	Compare simulated season results to the underlying talent levels of teams.	4, 6
22.4	Generalize the Monte Carlo framework to simulate other probabilistic events.	1, 3

Lesson 23 — The Great Streak (DiMaggio)

#	Lesson Objective	CO
23.1	Calculate run-length encoding (streaks) from a sequence of binary outcomes.	2
23.2	Compute a moving average to visualize streaky performance over a season.	2, 4
23.3	Use simulation (random shuffling) to assess how unusual DiMaggio's 56-game hitting streak was.	1, 3
23.4	Interpret p-values from a permutation/simulation test in the context of streak analysis.	4, 6

Lesson 24 — Streakiness Statistics

#	Lesson Objective	CO
24.1	Calculate a clumpiness/streakiness coefficient (sum of squared gap lengths) for any player.	3
24.2	Use simulation to generate the null distribution of the streakiness coefficient under randomness.	1, 3
24.3	Compare a player's observed streakiness to the simulated null distribution.	4, 6
24.4	Identify the longest hitless streaks among MLB players and interpret them.	1, 4

Lesson 26 — Statcast Spray Charts

#	Lesson Objective	CO
26.1	Retrieve Statcast data for individual batters using <code>scrape_statcast_savant()</code> .	2
26.2	Build a spray chart showing where batters hit balls in play on the field.	2, 4
26.3	Compare pull-happy vs. opposite-field hitters and relate tendencies to defensive shifts.	1, 4
26.4	Compute percentages of pulled ground balls and infield shifts faced for individual players.	2, 3

Lesson 27 — Launch Angles & Exit Velocity

#	Lesson Objective	CO
27.1	Create a scatter plot showing the relationship between exit velocity, launch angle, and batting average.	2, 4
27.2	Compare 2015 and 2017 Statcast data to investigate changes in home run rates.	1, 6
27.3	Use wOBA as an alternative outcome measure on launch-angle/exit-velocity plots.	3
27.4	Construct histograms of launch angle and exit velocity for home runs in different years.	2, 4

Lesson 28 — Home Run Hitting Part I

#	Lesson Objective	CO
28.1	Fit a GAM to model home run probability as a smooth function of launch speed and launch angle.	3
28.2	Use the 2015 model to predict home run probability on 2017 batted balls.	3, 6
28.3	Simulate 1,000 seasons to compare predicted vs. actual home run counts in 2017.	1, 3
28.4	Draw conclusions about whether changes in the baseball or changes in hitters better explain increased HR rates.	1, 5

Lesson 29 — Home Run Hitting Part II

#	Lesson Objective	CO
29.1	Estimate the optimal launch angle for hitting a home run using 2023 Statcast data.	1, 3
29.2	Analyze spray angle distributions for home runs by batter handedness.	2, 4
29.3	Compute park factors to determine which ballparks favor home run hitting.	3, 4

#	Lesson Objective	CO
29.4	Fit a random effects model to decompose home run responsibility between pitchers and batters.	3, 5

Lesson 30 — Equivalence Coefficient (Hall of Fame)

#	Lesson Objective	CO
30.1	Use logistic regression to model Hall of Fame selection probability from career statistics.	3
30.2	Compute the equivalence coefficient to project how a player's statistics would change with additional PA.	3
30.3	Compare Don Mattingly to Harold Baines and other Hall of Famers using raw and adjusted metrics.	1, 4
30.4	Critically evaluate whether the equivalence coefficient is a fair method for comparing players across different career lengths.	5, 6

Lesson 31 — OPS+ and BABIP

#	Lesson Objective	CO
31.1	Calculate OPS+ to compare batters across different offensive eras.	3, 4
31.2	Explain why raw OPS is not appropriate for cross-era comparisons and how OPS+ adjusts for league context.	5, 6
31.3	Compute Isolated Power (ISO) to distinguish between different types of hitters.	2, 3
31.4	Calculate BABIP and explain what it communicates about a player's performance on balls in play.	1, 4

Lesson 32 — Linear Weights

#	Lesson Objective	CO
32.1	Derive linear weights for singles, doubles, triples, home runs, walks, and outs from run-value data.	3
32.2	Compute Batting Runs for each player using Thorn and Palmer's linear weights model.	3
32.3	Compare Batting Runs to traditional statistics (AVG, SLG, OPS, RBI) and assess which better measures offensive contribution.	4, 6
32.4	Convert Batting Runs to wins using the 10-runs-per-win rule of thumb.	3, 4

Lesson 33 — Runs Created

#	Lesson Objective	CO
33.1	Compute the Basic, Stolen Base, and Technical versions of Runs Created from Lahman data.	2, 3
33.2	Diagnose data coverage limitations across baseball history for each Runs Created formula.	5, 6
33.3	Compute runs-created rate per game and use the Pythagorean formula to estimate a player's win percentage.	3
33.4	Use Runs Created to answer "Who is the best batter in history?" and "Which team had the greatest lineup?"	1, 4

Lesson 34 — Quarto Presentations

#	Lesson Objective	CO
34.1	Create a Quarto presentation (Reveal JS) that displays dynamic content from R code chunks.	2, 4
34.2	Use YAML parameters (<code>params</code>) to make a parameterized report that can be rendered for different inputs.	2

#	Lesson Objective	CO
34.3	Programmatically render multiple presentations in a loop, varying the parameter each time.	2
34.4	Scale analysis pipelines to produce customized products for any player, team, or season with minimal extra effort.	2, 5

Lesson 35 — Shiny Apps

#	Lesson Objective	CO
35.1	Build a basic Shiny app with a user interface (inputs) and server (reactive outputs).	2, 4
35.2	Connect slider inputs and radio button selections to dynamically rendered plots and tables.	2
35.3	Recognize the advantages (interactivity, customizability) and limitations (hosting, reproducibility) of Shiny compared to static documents.	5
35.4	Create a Shiny app that lets users explore batting statistics across user-selected time periods.	1, 2

Graded Events Summary

Problem Sets (8 total)

PS	Title	Lessons Covered	COs Assessed
1	Introduction to Course Datasets and Tidyverse Verbs	1–6	1, 2
2	Tidyverse Verbs and Graphics	2–5	1, 2
3	Runs and Bunts	6, 8, 9, 10	1, 2, 3, 5, 6
4	Balls, Strikes, and Umpire Variability	12, 13, 14, 15	1, 2, 3, 4, 5, 6
5	Mixed Models and Peak Trajectories	16, 18, 19, 20	1, 2, 3, 4, 5, 6

PS	Title	Lessons Covered	COs Assessed
6	Simulating Baseball Games with Markov Chains	21, 22	1, 2, 3, 4
7	Banning the Infield Shift	28, 29	1, 2, 3, 4, 5, 6
8	Understanding Sabermetrics	30, 31, 32, 33	1, 2, 3, 4

Exams

Exam	Lessons Covered	COs Assessed
WPR (Written Progress Review)	1–22	1, 3, 4, 5, 6
TEE (Term End Examination)	1–33	1, 3, 4, 5, 6

Course Project

Item	Lessons Covered	COs Assessed
SCORE Module	All	1, 2, 3, 4, 5, 6

Gap Analysis: Lesson Content vs. Graded Events

Well-Assessed Areas

- **Tidyverse data wrangling (Lsn 1–5):** PS1, PS2 comprehensively test these skills. ✓
- **Linear regression / Pythagorean model (Lsn 6, 8):** PS3, WPR Q1, TEE Q1 directly assess. ✓
- **Run expectancy matrix / run values (Lsn 9–10):** PS3, WPR Q2 assess interpretation and calculation. ✓
- **Logistic regression / GAMs (Lsn 14–15):** PS4, PS7, WPR Q3, TEE Q4 thoroughly assess. ✓
- **Random effects / multilevel models (Lsn 16, 20):** PS5 directly assesses mixed models. ✓
- **Career trajectories (Lsn 18–20):** PS5, TEE Q6 assess quadratic trajectories and multilevel shrinkage. ✓
- **Simulation (Lsn 21–24):** PS6, TEE Q3 assess Markov chains and permutation tests. ✓
- **Linear weights / OPS+ / equivalence coefficient (Lsn 30–33):** PS8, TEE Q7 assess these metrics. ✓
- **Communication / conclusions (All):** Project, PS discussion questions, and exam interpretations assess CO4. ✓

Partially Assessed Areas

Topic	Lessons	Assessment	Gap
String processing / regular expressions	Lsn 11	Not directly assessed	Students create count indicators in class but no graded event requires regex. Consider adding a regex question to a PS or exam.
LOESS models	Lsn 12	Not directly assessed	LOESS is used to model umpire strike zones but only GAMs appear on graded events. Minor gap — GAMs subsume this.
Bayesian multilevel models (brms)	Lsn 20	PS5 covers multilevel concepts, but the Bayesian fitting (brms/MCMC) is not assessed	Acceptable since PS5 covers the key concept (shrinkage) through <code>lme4</code> instead.
Spray charts	Lsn 26	Not directly assessed	PS7 touches on shift analysis using GAMs, but constructing spray charts is not tested. Consider a PS question.
Launch angle / exit velocity visualization	Lsn 27	TEE Q5 examines the conceptual understanding of the GAM surface	Reasonable coverage.
Home run simulation (2015→2017)	Lsn 28	PS7 mirrors this approach for the shift analysis	Adequately covered by transfer of the method.
Park factors	Lsn 29	TEE Q7 asks for a definition only	Could be strengthened with a computation question.
BABIP	Lsn 31	Not directly assessed	Simple calculation — could be added to PS8 or an exam.
Runs Created formulas	Lsn 33	PS8 does not cover this; the “solution” file for PS8 includes it but the actual PS8 omits it	The solution <code>.qmd</code> contains RC analysis but the student-facing PS8 does not. This is a gap — consider adding an RC problem.

Not Assessed Areas

Topic	Lessons	Why It Matters
Quarto presentations (parameterized reporting)	Lsn 34	This lesson supports the project presentation component, but is not itself graded. Students are not required to demonstrate parameterized rendering. If this skill is a course goal, consider a small graded deliverable.
Shiny apps	Lsn 35	This is an enrichment lesson. Students build a Shiny app as an in-class exercise but it is not graded. This is acceptable if kept as enrichment.

Course Objective Coverage Summary

CO	PS1	PS2	PS3	PS4	PS5	PS6	PS7	PS8	WPR	TEE	Project
CO1 (Investigate)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO2 (Software)	✓	✓	✓	✓	✓	✓	✓	✓	—	—	✓
CO3 (Methods)	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO4 (Communicate)	—	—	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO5 (Limitations)	—	—	✓	✓	✓	—	✓	—	✓	✓	✓
CO6 (Critical consumer)	—	—	✓	✓	✓	—	✓	—	✓	✓	✓

Recommended Adjustments

1. **PS8 should include a Runs Created problem.** The student-facing PS8 only covers equivalence coefficient, OPS+, and linear weights, but the solution `.qmd (lsn_33_runs_created_solution.qmd)` includes a full RC analysis. This is a disconnect.
2. **Add a string processing question to a problem set or exam.** Regular expressions for pitch sequence data (Lsn 11) is a valuable data science skill that is currently untested.

3. **CO5 is lightly assessed in PS3 and PS8.** The project is the main vehicle for CO5. Consider adding a brief reflection question to PS6 (simulation) asking students to discuss the limitations of the Markov assumption.
4. **CO2 is not assessed by WPR or TEE** (both are written exams with no coding). This is by design — the project covers CO2 comprehensively. No change needed.
5. **PS6 does not assess CO5 or CO6.** Consider adding a question about the limitations of the Markov chain memoryless property in the context of baseball momentum (already a lesson objective 21.4).